

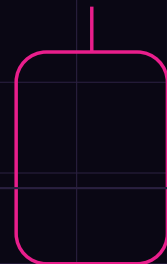
AUTOMATED BIOREACTOR ENVIRONMENTAL CONTROL UNIT

for Pharmaceutical Batch Processing

TECHNICAL PROJECT MANUAL · REV. A

By Sipho Lucky Sibanda

Platform: Siemens S7-1500 (SCL) / CODESYS-portable Structured Text
Simulation & Portfolio Engineering Build · Not for GMP Production Use



BR-201



FRONT MATTER

Document Control & Disclaimer

This document is a self-authored technical project manual produced as part of a personal engineering portfolio. It describes the design, control philosophy, and simulated validation of a bioreactor environmental control system, built to demonstrate ISA-88 batch sequencing, cascade and split-range process control, and sterilisation-validation mathematics for automation, controls, and biotech/pharma engineering roles.

The system described here was developed and tested in simulation only (PLCSIM-style forcing of inputs and desktop review). No part of this project has been installed, commissioned, or verified on physical bioprocess hardware, and it must not be treated as a validated GMP manufacturing system.

HONEST LIMITATION — Portfolio project, not a validated GMP system

A real biopharma bioreactor control system requires full process validation, 21 CFR Part 11 compliant electronic batch records, and qualification against the specific cell line and product involved, none of which this project claims. Figures, thresholds, and I/O in this manual are engineering-realistic but illustrative.

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FRONT MATTER

Executive Summary

This is the fourth deliberate discipline shift in the wider automation portfolio, moving from mechanical/electrical process control toward validated pharmaceutical batch manufacturing - a field where the control logic and the regulatory record it produces are equally important. Producing a biologic drug substance means holding a living cell culture inside a precise, validated environment for days, after first proving the vessel itself was sterilised beyond reasonable doubt.

FB_Bioreactor_BatchControl sequences a full batch through the ISA-88 style phases a real biopharma batch record follows: CIP, SIP (sterilisation), Cooldown, Inoculation, Incubation, and Harvest. Sterilisation itself is validated using the real F0 lethality formula rather than a fixed timer, because sterilisation is temperature and time dependent - holding 110°C for an hour is not equivalent to 121°C for six minutes, and F0 is the industry-standard way of comparing the two on one scale.

Two linked control loops keep the culture environment in spec throughout incubation: a cascade temperature loop (vessel temperature setting a jacket temperature target, which itself drives the heating/cooling valves) and a split-range dissolved-oxygen loop that exhausts agitation authority before ever opening an oxygen sparge valve. pH is held with deadband on/off dosing rather than a continuous controller, avoiding pump chatter on a naturally noisy signal.

What follows documents the regulatory and industry context, architecture, hardware assumptions, full I/O list and recipe structure, control philosophy, complete annotated code, HMI design, alarm philosophy, and the functional test procedure used to validate the logic in simulation.

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BIOREACTOR CONTROL

Industry & Regulatory Context

Biopharmaceutical manufacturing operates under Good Manufacturing Practice (GMP), enforced by regulators including the FDA and EMA. Every batch's process parameters, deviations, and outcomes must be documented in a way that can withstand regulatory audit - the control system isn't just running the process, it's generating the record that proves the process was run correctly.

ISA-88: the standard structure behind this project

ISA-88 (IEC 61512) is the batch-control standard this project's phase structure deliberately mirrors: a recipe defines the parameters for a specific product, a procedure breaks the batch into ordered phases, and each phase runs its own permissive-gated logic before handing control to the next. This project's `ST_BatchRecipe` structure and its CIP/SIP/Cooldown/Inoculation/Incubation/Harvest phase sequence are a simplified but structurally faithful implementation of that model.

Why sterilisation validation is its own discipline

Steam sterilisation (SIP - Sterilise In Place) doesn't just need to reach a target temperature; it needs to prove enough cumulative thermal lethality was delivered to kill the resistant spore-forming organisms process validation is built around. That proof is what F0 represents, and it is a real, standard part of sterilisation validation across pharma and food processing alike, not a concept invented for this project.

NOTE — The batch record is the product, almost as much as the drug substance is

In a GMP environment, a batch with no F0 record, or an ambiguous one, cannot be released regardless of how the culture actually grew - this is why Chapter 5's insistence on treating an aborted, under-sterilised SIP as an explicit failure (rather than silently proceeding) reflects real regulatory stakes, not an abstract design preference.

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BIOREACTOR CONTROL

System Architecture & Batch Overview

A single stirred-tank bioreactor vessel is fitted with a heating/cooling jacket, a variable-speed agitator, an oxygen sparge line, acid/base dosing pumps, and pH/DO/temperature/optical-density probes. A SCADA/MES layer downloads the batch recipe; the PLC owns every control decision from CIP through harvest.

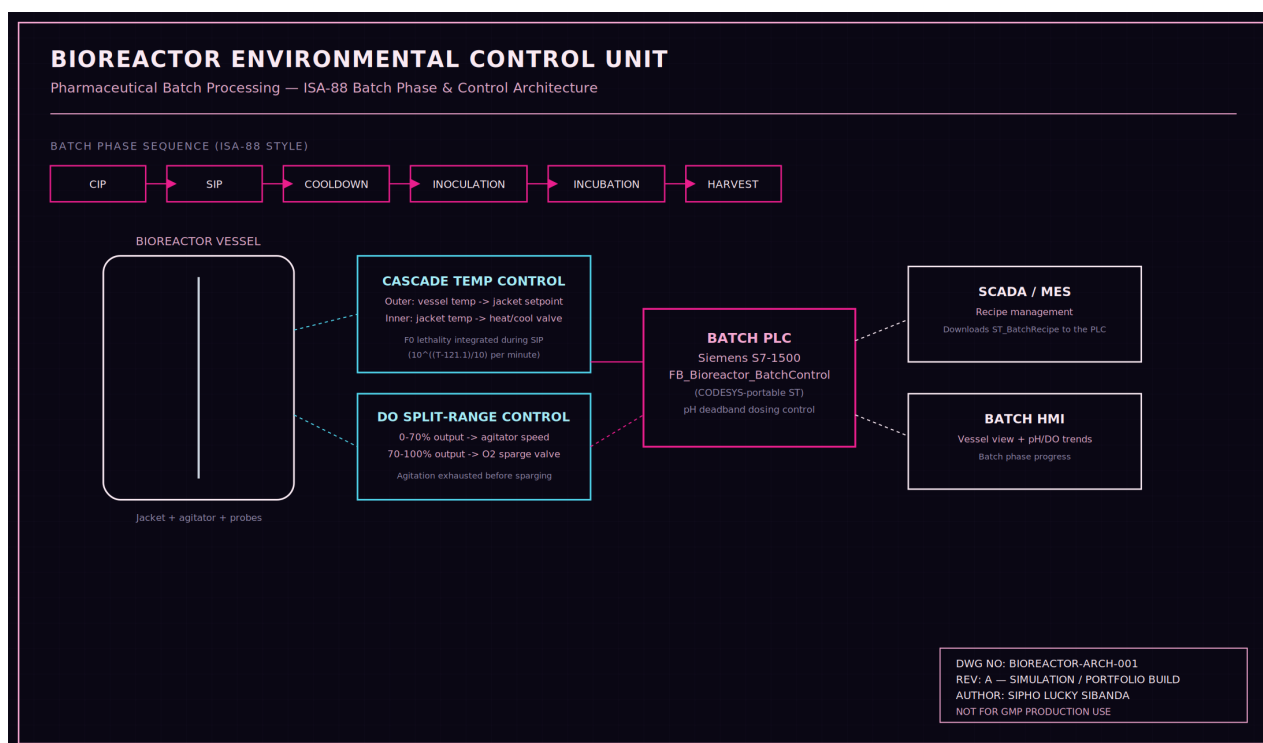


Figure 2.1 — Batch phase sequence and control architecture. Dashed lines represent PLC signal and configuration paths.

Control hierarchy

- Field layer — temperature, DO, pH, and optical density probes; heating/cooling valves; variable-speed agitator; oxygen sparge valve; acid/base dosing pumps.
- Control layer — a single PLC function block, FB_Bioreactor_BatchControl, hosted on a Siemens S7-1500 (or CODESYS-based pharmaceutical process controller), running the full batch sequence and every control loop.
- Supervisory layer — SCADA/MES recipe management (upstream, downloading the recipe) and the batch HMI (downstream, presenting vessel state and trends).

Why the recipe is an input structure, not embedded constants

Passing Recipe : ST_BatchRecipe into the function block as a whole structure - rather than hardcoding setpoints - mirrors real ISA-88 practice: the same equipment module runs many products over its lifetime, each with its own validated recipe, without the control logic itself ever changing.

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BIOREACTOR CONTROL

Hardware & Instrumentation Specification

As with the rest of this portfolio, the logic here is platform-portable but written against a concrete reference platform so thresholds and timing are grounded.

Component	Representative Spec	Role
Batch PLC CPU	Siemens SIMATIC S7-1500	Executes FB_Bioreactor_BatchControl
Vessel/Jacket RTD	Pt100, 4-20mA, 0-140C	Cascade temperature control feedback
DO Probe	Polarographic or optical, 4-20mA, 0-100%	Split-range DO control feedback
pH Probe	Glass electrode, 4-20mA, 0-14	Deadband dosing control feedback
Optical Density Probe	In-situ OD600-equivalent sensor	Early-harvest trigger
Variable-Speed Agitator	0-500 RPM, 4-20mA speed reference	Mixing / DO split-range primary
Oxygen Sparge Valve	4-20mA proportional control valve	DO split-range secondary
Acid / Base Dosing Pumps	24VDC on/off metering pumps	pH deadband control

DESIGN NOTE — Why deadband dosing, not a continuous pH controller

A pH probe signal is naturally noisy at the resolution a dosing decision needs. A continuous PID output driving an on/off pump would chatter the pump open and closed on every small fluctuation, wasting reagent and potentially destabilising pH more than it stabilises it. A fixed deadband around setpoint is the standard, deliberately simple answer used throughout real bioprocess control.

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BIOREACTOR CONTROL

I/O List & Recipe Parameters

The table below is the working I/O list for FB_Bioreactor_BatchControl (see also docs/IO_List.md in the repository, and Appendix A of this manual).

Recipe parameters (downloaded from SCADA/MES)

Field	Typical Value
SIP_Temp_Setpoint_C	121.1 °C
SIP_Target_F0_min	15.0 min
Cooldown_Temp_C	37.0 °C
Incubation_Temp_Setpoint_C	37.0 °C
Incubation_DO_Setpoint_Pct	40.0 %
Incubation_pH_Setpoint	7.00
Incubation_Duration_hr	72.0 hr
Harvest_OD_Threshold	8.0 OD

Key inputs and outputs

Tag	Description	Signal
AI_VesselTemp_C / AI_JacketTemp_C	Cascade loop feedback	4-20mA x2
AI_DO_Pct / AI_pH / AI_OpticalDensity	Process measurements	4-20mA x3
DI_CIP_Complete / DI_StartBatch / DI_AbortBatch	Batch sequencing commands	Digital x3
AO_HeatingValve_Pct / AO_CoolingValve_Pct	Jacket control outputs	4-20mA x2
AO_AgitatorSpeed_RPM / AO_O2SpargeValve_Pct	DO split-range outputs	4-20mA x2
DO_AcidPump / DO_BasePump	pH dosing outputs	Digital x2
BatchPhase / Current_F0_min / SystemStatus	State & batch record data	Mixed

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BIOREACTOR CONTROL

Control Philosophy: Cascade, Split-Range & F0 Lethality

Three distinct control techniques do the real work in this project, each solving a genuinely different kind of problem.

5.1 Cascade control: why two loops beat one

A single PID loop reading vessel temperature and driving the jacket valve directly would work, eventually - but slowly, because heat has to move from the jacket, through the vessel wall, into the bulk liquid before the controller ever sees the effect of its own action. Cascade control splits this into two loops: the outer loop compares vessel temperature to setpoint and outputs a jacket temperature target; the inner loop compares actual jacket temperature to that target and drives the valves directly. The inner loop reacts to a heat-transfer-fluid disturbance in seconds, long before it would ever show up as a vessel temperature deviation.

5.2 Split-range control: agitation before oxygen

Dissolved oxygen can be raised two ways: spin the impeller faster (cheap, fast, but limited by shear stress on the cells at high speed) or bubble more oxygen through the broth (effective, but oxygen is a real cost and over-sparging can strip out other dissolved gases). This project's split-range logic spends the first 70% of the DO controller's output entirely on agitation, and only opens the oxygen sparge valve once agitation is already at its ceiling.

DESIGN NOTE — The split point (70%) is a design choice, and it's named as one

There's nothing physically fundamental about 70% specifically - it's a tuning decision balancing shear sensitivity against oxygen cost for a given cell line and vessel geometry. A real commissioning engineer would set this based on the actual organism and vessel being used; this project states the number plainly as a parameter rather than presenting it as a universal constant.

5.3 F0: comparing sterilisation across temperature and time together

The standard formula is:

$$F_0 = \sum 10^{((T - 121.1) / z) \times \Delta t} \quad (z = 10)$$

Every 10°C above the 121.1°C reference temperature multiplies the lethality rate by ten; every 10°C below divides it by ten. Integrating that rate over the actual temperature profile the vessel experienced - not just the setpoint - produces a single number, in equivalent minutes at 121.1°C, that can be compared directly against a validated target.

This is exactly why SIP completes on accumulated F0, not on elapsed time: a vessel that ran slightly cool for part of the hold needs longer to reach the same lethality, and F0-based completion accounts for that automatically.

5.4 A failure during SIP must never look like success

If a batch is aborted mid-SIP, Alarm_SIP_Failed is raised whenever the accumulated F0 fell short of the recipe's target. This exists so a batch record can never ambiguously suggest a vessel was safe to inoculate when the sterilisation target was never actually reached - a direct consequence of Chapter 1's point that the record is almost as important as the process itself.

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BIOREACTOR CONTROL

PLC Logic Walkthrough

This chapter walks through FB_Bioreactor_BatchControl section by section. The full listing is reproduced in Appendix B.

6.1 The cascade loop

```
OuterTemp_PID(SP := Recipe.Incubation_Temp_Setpoint_C, PV := AI_VesselTemp_C,
              KP := OuterTemp_Kp, TI := OuterTemp_Ti, MANUAL := FALSE,
              LMN_HLM := 135.0, LMN_LLM := 0.0);
AO_JacketTempSetpoint_C := OuterTemp_PID.LMN;

InnerJacketTemp_PID(SP := AO_JacketTempSetpoint_C, PV := AI_JacketTemp_C,
                   KP := InnerJacket_Kp, TI := InnerJacket_Ti, MANUAL := FALSE,
                   LMN_HLM := 100.0, LMN_LLM := -100.0);

IF InnerJacketTemp_PID.LMN >= 0.0 THEN
    AO_HeatingValve_Pct := InnerJacketTemp_PID.LMN;
ELSE
    AO_CoolingValve_Pct := ABS(InnerJacketTemp_PID.LMN);
END_IF
```

Listing 6.1 — The outer loop's output literally becomes the inner loop's setpoint.

6.2 F0 accumulation

```
T_ScanTime(IN := (BatchPhase = PH_SIP), PT := T_ScanTime_PT);
IF BatchPhase = PH_SIP THEN
    IF T_ScanTime.Q THEN
        T_ScanTime(IN := FALSE);
        LethalityRate_PerMin := EXPT(10.0, (AI_VesselTemp_C - F0_RefTemp_C) / F0_zValue);
        Current_F0_min := Current_F0_min + (LethalityRate_PerMin * (1.0 / 60.0));
    END_IF
ELSIF BatchPhase = PH_CIP THEN
    Current_F0_min := 0.0;
END_IF
```

Listing 6.2 — Lethality is integrated once per second against the ACTUAL vessel temperature.

6.3 Split-range DO control

```
IF DO_ControllerOutput_Pct <= 70.0 THEN
    AO_AgitatorSpeed_RPM := Agitator_Min_RPM
        + (DO_ControllerOutput_Pct / 70.0) * (Agitator_Max_RPM - Agitator_Min_RPM);
    AO_O2SpargeValve_Pct := 0.0;
ELSE
    AO_AgitatorSpeed_RPM := Agitator_Max_RPM;
    AO_O2SpargeValve_Pct := (DO_ControllerOutput_Pct - 70.0) / 30.0 * 100.0;
END_IF
```

Listing 6.3 — Agitation authority is fully spent before sparge ever opens.

6.4 SIP completion and failure flagging

```
PH_SIP:
  IF Current_F0_min >= Recipe.SIP_Target_F0_min THEN
    BatchPhase := PH_COOLDOWN;
  END_IF

// ... elsewhere, evaluated every scan ...
Alarm_SIP_Failed := (BatchPhase = PH_ABORT) AND (Current_F0_min < Recipe.SIP_Target_F0_min);
```

Listing 6.4 — SIP advances on lethality, not time; an abort short of target is flagged, not hidden.

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BIOREACTOR CONTROL

HMI Design & Live Trends

The HMI mockup (hmi/index.html in the repository) centres on a vessel cross-section - jacket, agitator, liquid fill, rising bubbles when sparging - paired with scrolling trend charts for pH, dissolved oxygen, and temperature, exactly the operator view a real batch record review would reference.

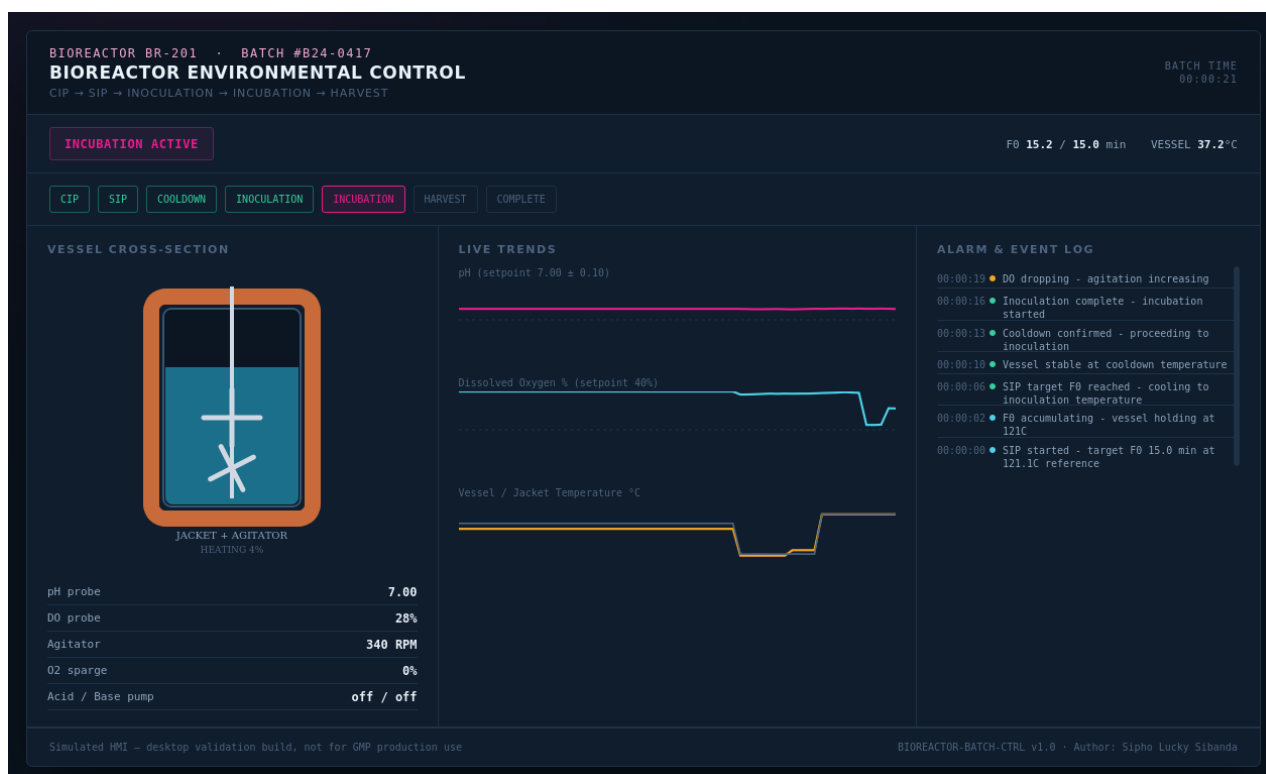


Figure 7.1 — Incubation active: agitator spinning, jacket in a light heating trim, live trends showing the cooldown transition and the DO recovery curve from a scripted split-range event.

Design decisions

- Magenta/cyan "cleanroom" palette — a fifth distinct visual identity in this portfolio, chosen to read as clinical/biotech rather than heavy industrial, appropriate for a process governed by GMP rather than a switchboard or safety panel.
- The phase chip row is always visible — CIP through Complete, with the active phase highlighted, so an operator glancing at the screen always knows exactly where in the batch lifecycle the vessel currently sits.
- Bubbles only appear when sparge is actually active — a small detail, but one that ties the visual directly to the split-range logic from Chapter 5 rather than running as generic decoration.
- Real scrolling trends, not static gauges — matching the recruiter-facing brief this project was built against: a chemical vat simulation with real-time graphs tracking pH and dissolved oxygen, not just instantaneous numbers.

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BIOREACTOR CONTROL

Alarm Philosophy & Fail-Safe Design

Every alarm in this system maps to a specific batch-record-relevant condition, consistent with the rest of this portfolio.

Condition	Meaning	Expected Response
Alarm_SIP_Failed	Batch aborted before F0 target reached	Vessel must be re-sterilised - never inoculate on an unproven SIP
Alarm_DO_Low	Dissolved oxygen below 50% of setpoint	Check sparge/agitation authority; investigate a possible sensor or supply fault
Alarm_pH_Excursion	pH more than 0.5 from setpoint	Investigate dosing pump function and reagent supply
Alarm_TempDeviation	Vessel temperature more than 2°C from setpoint	Investigate cascade loop tuning or a jacket-side fault

Fail-safe defaults, summarised

- Batch abort → cooling valve to 100%, agitator/sparge/dosing all off - pulling heat out and stopping every active input to the vessel.
- An aborted SIP short of its F0 target is explicitly flagged, never silently treated as equivalent to a completed one.
- pH dosing pumps default off outside the Incubation phase - no dosing action is ever possible during CIP, SIP, or cooldown.

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BIOREACTOR CONTROL

Testing, Commissioning & FAT Procedures

The function block was validated against fourteen functional test cases, anchored by two F0-accumulation-rate tests with known, calculable correct answers. The full procedure is in docs/Testing_Procedures.md; the matrix is reproduced below.

#	Test Case	Expected Result
1-2	Batch start / CIP handoff to SIP	Phase transitions correctly; F0 resets at CIP entry
3	F0 rate at reference temperature	~1.0 min/min accumulation at exactly 121.1C
4	F0 rate 10C above reference	10x faster accumulation than Test 3
5	SIP completes on F0, not time	Transitions on accumulated lethality alone
6	Cascade response to a temperature step	Jacket setpoint and heating valve respond correctly
7	Cooldown requires a stable hold	Confirmed only after a full 5-minute settle, not on first touch
8-9	DO split-range, both regions	Agitation-only below 70% output; sparge engages above it
10-11	pH deadband and excursion alarm	No dosing chatter within deadband; alarm past 0.5 deviation
12	Early harvest on optical density	Transitions to Harvest before the incubation timer expires
13-14	Abort safe state and SIP failure flag	Full safe-state outputs; F0 shortfall explicitly flagged

ENGINEERING NOTE — Why Tests 3 and 4 anchor the whole suite

F0 accumulation rate at the reference temperature has a known correct answer (approximately 1.0 equivalent minute per real minute), and the rate exactly 10 degrees above reference must be exactly ten times faster by the formula's own definition. If either of these doesn't land, no SIP completion decision anywhere downstream can be trusted, regardless of how reasonable it looks in isolation.

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BIOREACTOR CONTROL

Limitations, Real-World Deltas & Future Work

Naming the gap between a strong simulation and a validated GMP system is part of the engineering, consistent with every other project in this portfolio.

What a real installation would add

- 21 CFR Part 11 compliant electronic batch records — audit trails, electronic signatures, and data integrity controls this project makes no claim to.
- A maximum SIP duration alarm — named directly in the code comments (Chapter 6.4): an unreachable F0 target should never be able to hold a vessel at high temperature indefinitely without operator intervention.
- A fully sequenced inoculation transfer — this project's Inoculation phase advances on a settle timer; a real recipe would sequence valve alignments, a transfer pump, and a confirmed volume addition.
- Process validation — qualification runs proving the control strategy actually holds spec for the specific cell line and product involved.

10.1 Hazard & safeguard register (HAZOP-style)

Hazard	Cause	Safeguard in This Design
Under-sterilised vessel inoculated	SIP aborted or ended prematurely	F0-based completion, not a timer; Alarm_SIP_Failed on any shortfall
Cell culture shear damage	Agitation pushed too high chasing DO alone	Split-range control shares the DO response with oxygen sparging
Runaway high SIP temperature	No maximum SIP duration/temperature ceiling modelled	Named gap - a real system needs an independent high-temperature trip
Dosing pump reagent depletion undetected	No reagent level/flow feedback in this I/O set	Named gap - a real system would add tank level monitoring on both dosing lines

Where this project could go next

- Add an independent, hard-wired high-temperature trip during SIP, separate from the F0-based software logic.
- Model a second vessel and a shared CIP/SIP skid, demonstrating equipment-module arbitration per the full ISA-88 model.
- Extend the batch record output to a structured, timestamped export suitable for a real electronic batch record system to ingest.

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APPENDIX

Appendix A — I/O Quick Reference

Tag	Dir.	Type	Notes
Recipe	IN	ST_BatchRecipe	From SCADA/MES
AI_VesselTemp_C / AI_JacketTemp_C	IN	REAL x2	Cascade feedback
AI_DO_Pct / AI_pH / AI_OpticalDensity	IN	REAL x3	Process measurements
AI_AgitatorSpeed_RPM	IN	REAL	Feedback
DI_CIP_Complete / StartBatch / AbortBatch	IN	BOOL x3	Sequencing
DI_System_Enable	IN	BOOL	Master enable
AO_JacketTempSetpoint_C	OUT	REAL	Cascade intermediate
AO_HeatingValve_Pct / CoolingValve_Pct	OUT	REAL x2	Jacket control
AO_AgitatorSpeed_RPM / O2SpargeValve_Pct	OUT	REAL x2	DO split-range
DO_AcidPump / DO_BasePump	OUT	BOOL x2	pH deadband dosing
BatchPhase	OUT	ENUM	9-state machine
Current_F0_min / BatchElapsed_hr	OUT	REAL x2	Batch record data
Alarm_SIP_Failed / DO_Low / pH_Excursion / TempDeviation	OUT	BOOL x4	Process alarms
SystemStatus	OUT	STRING	HMI display

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APPENDIX

Appendix B — Full Structured Text Listing

Complete, unedited listing of src/Bioreactor_BatchControl.st.

```
(*
=====
PROJECT : Automated Bioreactor Environmental Control Unit
          for Pharmaceutical Batch Processing
MODULE   : FB_Bioreactor_BatchControl
PLATFORM : IEC 61131-3 Structured Text (Siemens SCL / CODESYS-portable)
AUTHOR   : Sipho Lucky Sibanda
          See Chapter 5 for cascade, split-range, and F0 lethality discussion.
=====
*)

TYPE E_BatchPhase :
(
    PH_IDLE, PH_CIP, PH_SIP, PH_COOLDOWN, PH_INOCULATION,
    PH_INCUBATION, PH_HARVEST, PH_COMPLETE, PH_ABORT
);
END_TYPE

TYPE ST_BatchRecipe :
STRUCT
    SIP_Temp_Setpoint_C : REAL; SIP_Target_F0_min : REAL;
    Cooldown_Temp_C : REAL;
    Incubation_Temp_Setpoint_C : REAL; Incubation_DO_Setpoint_Pct : REAL;
    Incubation_pH_Setpoint : REAL; Incubation_Duration_hr : REAL;
    Harvest_OD_Threshold : REAL;
END_STRUCT
END_TYPE

FUNCTION_BLOCK FB_Bioreactor_BatchControl
VAR_INPUT
    Recipe : ST_BatchRecipe;
    AI_VesselTemp_C : REAL; AI_JacketTemp_C : REAL;
    AI_DO_Pct : REAL; AI_pH : REAL;
    AI_OpticalDensity : REAL; AI_AgitatorSpeed_RPM : REAL;
    DI_CIP_Complete : BOOL; DI_StartBatch : BOOL;
    DI_AbortBatch : BOOL; DI_System_Enable : BOOL;
END_VAR
```

```

VAR_OUTPUT
  A0_JacketTempSetpoint_C : REAL := 0.0;
  A0_HeatingValve_Pct : REAL := 0.0; A0_CoolingValve_Pct : REAL := 0.0;
  A0_AgitatorSpeed_RPM : REAL := 0.0; A0_O2SpargeValve_Pct : REAL := 0.0;
  DO_AcidPump : BOOL := FALSE; DO_BasePump : BOOL := FALSE;
  BatchPhase : E_BatchPhase := PH_IDLE;
  Current_F0_min : REAL := 0.0; BatchElapsed_hr : REAL := 0.0;
  Alarm_SIP_Failed : BOOL := FALSE; Alarm_DO_Low : BOOL := FALSE;
  Alarm_pH_Excursion : BOOL := FALSE; Alarm_TempDeviation : BOOL := FALSE;
  SystemStatus : STRING[28] := 'STANDBY';
END_VAR

VAR
  OuterTemp_PID : FB_PID; InnerJacketTemp_PID : FB_PID;
  OuterTemp_Kp : REAL := 3.0; OuterTemp_Ti : REAL := 300.0;
  InnerJacket_Kp : REAL := 6.0; InnerJacket_Ti : REAL := 60.0;

  DO_PID : FB_PID;
  DO_Kp : REAL := 4.0; DO_Ti : REAL := 120.0;
  DO_ControllerOutput_Pct : REAL;
  Agitator_Min_RPM : REAL := 100.0; Agitator_Max_RPM : REAL := 400.0;

  pH_Deadband : REAL := 0.10;

  T_ScanTime : TON; T_ScanTime_PT : TIME := T#1S;
  F0_RefTemp_C : REAL := 121.1; F0_zValue : REAL := 10.0;
  LethalityRate_PerMin : REAL;

  T_IncubationTimer : TON;
  T_CooldownConfirm : TON; T_CooldownConfirm_PT : TIME := T#5M;
  TempDeviation_Deg : REAL := 2.0;
END_VAR

```

```
// 1. CASCADE TEMPERATURE CONTROL
IF (BatchPhase = PH_SIP) OR (BatchPhase = PH_COOLDOWN)
  OR (BatchPhase = PH_INOCULATION) OR (BatchPhase = PH_INCUBATION) THEN
  CASE BatchPhase OF
    PH_SIP: OuterTemp_PID(SP := Recipe.SIP_Temp_Setpoint_C, PV := AI_VesselTemp_C,
      KP := OuterTemp_Kp, TI := OuterTemp_Ti, MANUAL := FALSE, LMN_HLM := 135.0,
      LMN_LLM := 0.0);
    PH_COOLDOWN: OuterTemp_PID(SP := Recipe.Cooldown_Temp_C, PV := AI_VesselTemp_C,
      KP := OuterTemp_Kp, TI := OuterTemp_Ti, MANUAL := FALSE, LMN_HLM := 135.0,
      LMN_LLM := 0.0);
    PH_INOCULATION, PH_INCUBATION: OuterTemp_PID(SP := Recipe.Incubation_Temp_Setpoint_C, PV :=
      AI_VesselTemp_C,
      KP := OuterTemp_Kp, TI := OuterTemp_Ti, MANUAL := FALSE, LMN_HLM := 135.0,
      LMN_LLM := 0.0);
  ELSE ;
  END_CASE
  AO_JacketTempSetpoint_C := OuterTemp_PID.LMN;
  InnerJacketTemp_PID(SP := AO_JacketTempSetpoint_C, PV := AI_JacketTemp_C,
    KP := InnerJacket_Kp, TI := InnerJacket_Ti, MANUAL := FALSE, LMN_HLM := 100.0, LMN_LLM :=
    -100.0);
  IF InnerJacketTemp_PID.LMN >= 0.0 THEN
    AO_HeatingValve_Pct := InnerJacketTemp_PID.LMN; AO_CoolingValve_Pct := 0.0;
  ELSE
    AO_HeatingValve_Pct := 0.0; AO_CoolingValve_Pct := ABS(InnerJacketTemp_PID.LMN);
  END_IF
  Alarm_TempDeviation := ABS(AI_VesselTemp_C - OuterTemp_PID.SP) > TempDeviation_Deg;
ELSE
  AO_HeatingValve_Pct := 0.0; AO_CoolingValve_Pct := 0.0; Alarm_TempDeviation := FALSE;
END_IF

// 2. F0 LETHALITY ACCUMULATION
T_ScanTime(IN := (BatchPhase = PH_SIP), PT := T_ScanTime_PT);
IF BatchPhase = PH_SIP THEN
  IF T_ScanTime.Q THEN
    T_ScanTime(IN := FALSE);
    LethalityRate_PerMin := EXPT(10.0, (AI_VesselTemp_C - F0_RefTemp_C) / F0_zValue);
    Current_F0_min := Current_F0_min + (LethalityRate_PerMin * (1.0 / 60.0));
  END_IF
ELSIF BatchPhase = PH_CIP THEN
  Current_F0_min := 0.0;
END_IF
```

```
// 3. DO SPLIT-RANGE CONTROL
IF BatchPhase = PH_INCUBATION THEN
    DO_PID(SP := Recipe.Incubation_DO_Setpoint_Pct, PV := AI_DO_Pct,
           KP := DO_Kp, TI := DO_Ti, MANUAL := FALSE, LMN_HLM := 100.0, LMN_LLM := 0.0);
    DO_ControllerOutput_Pct := DO_PID.LMN;
    IF DO_ControllerOutput_Pct <= 70.0 THEN
        AO_AgitatorSpeed_RPM := Agitator_Min_RPM + (DO_ControllerOutput_Pct / 70.0) *
(Agitator_Max_RPM - Agitator_Min_RPM);
        AO_02SpargeValve_Pct := 0.0;
    ELSE
        AO_AgitatorSpeed_RPM := Agitator_Max_RPM;
        AO_02SpargeValve_Pct := (DO_ControllerOutput_Pct - 70.0) / 30.0 * 100.0;
    END_IF
    Alarm_DO_Low := AI_DO_Pct < (Recipe.Incubation_DO_Setpoint_Pct * 0.5);
ELSE
    AO_AgitatorSpeed_RPM := 0.0; AO_02SpargeValve_Pct := 0.0; Alarm_DO_Low := FALSE;
END_IF

// 4. pH DEADBAND DOSING
IF BatchPhase = PH_INCUBATION THEN
    DO_AcidPump := AI_pH > (Recipe.Incubation_pH_Setpoint + pH_Deadband);
    DO_BasePump := AI_pH < (Recipe.Incubation_pH_Setpoint - pH_Deadband);
    Alarm_pH_Excursion := ABS(AI_pH - Recipe.Incubation_pH_Setpoint) > 0.5;
ELSE
    DO_AcidPump := FALSE; DO_BasePump := FALSE; Alarm_pH_Excursion := FALSE;
END_IF
```

```

// 5. BATCH PHASE STATE MACHINE
IF DI_AbortBatch THEN BatchPhase := PH_ABORT; END_IF
CASE BatchPhase OF
  PH_IDLE: IF DI_System_Enable AND DI_StartBatch THEN BatchPhase := PH_CIP; END_IF
  PH_CIP: IF DI_CIP_Complete THEN BatchPhase := PH_SIP; END_IF
  PH_SIP: IF Current_F0_min >= Recipe.SIP_Target_F0_min THEN BatchPhase := PH_COOLDOWN; END_IF
  PH_COOLDOWN:
    T_CooldownConfirm(IN := ABS(AI_VesselTemp_C - Recipe.Cooldown_Temp_C) < 1.0, PT :=
T_CooldownConfirm_PT);
    IF T_CooldownConfirm.Q THEN BatchPhase := PH_INOCULATION; END_IF
  PH_INOCULATION:
    T_IncubationTimer(IN := TRUE, PT := T#2M);
    IF T_IncubationTimer.Q THEN T_IncubationTimer(IN := FALSE); BatchPhase := PH_INCUBATION;
END_IF
  PH_INCUBATION:
    T_IncubationTimer(IN := TRUE, PT := REAL_TO_TIME(Recipe.Incubation_Duration_hr *
3600000.0));
    BatchElapsed_hr := TIME_TO_REAL(T_IncubationTimer.ET) / 3600000.0;
    IF T_IncubationTimer.Q OR (AI_OpticalDensity >= Recipe.Harvest_OD_Threshold) THEN
      T_IncubationTimer(IN := FALSE); BatchPhase := PH_HARVEST;
    END_IF
  PH_HARVEST: BatchPhase := PH_COMPLETE;
  PH_COMPLETE: ;
  PH_ABORT:
    AO_HeatingValve_Pct := 0.0; AO_CoolingValve_Pct := 100.0;
    AO_AgitatorSpeed_RPM := 0.0; AO_O2SpargeValve_Pct := 0.0;
    DO_AcidPump := FALSE; DO_BasePump := FALSE;
  ELSE ;
END_CASE
Alarm_SIP_Failed := (BatchPhase = PH_ABORT) AND (Current_F0_min < Recipe.SIP_Target_F0_min);

END_FUNCTION_BLOCK

```

Listing B.1 — Complete FB_Bioreactor_BatchControl source.

OC

APPENDIX

Appendix C — Glossary

Term	Meaning
ISA-88	The standard batch-control model (recipes, phases, procedures) this project follows
GMP	Good Manufacturing Practice — the regulatory framework for pharma manufacturing
CIP / SIP	Clean-in-Place / Sterilise-in-Place — automated vessel cleaning and sterilisation
F0	Accumulated sterilisation lethality, expressed as equivalent minutes at 121.1C
Cascade control	An outer loop's output becomes an inner loop's setpoint
Split-range control	One controller output driving two final elements over different output ranges
Dissolved Oxygen (DO)	The concentration of oxygen available to the culture, expressed as % of saturation
OD / OD600	Optical Density — a common proxy measurement for cell culture density
Deadband control	An on/off control strategy with a tolerance band to prevent output chatter
Batch record	The documented history of a specific manufacturing batch, required for GMP release



CLOSING

About the Author

Sipho Lucky Sibanda is an automation and controls engineer building a multi-disciplinary portfolio spanning marine systems, avionics, architectural technology, applied AI, biotechnology, and industrial automation. This manual documents the fourth deliberate discipline shift in that portfolio, following the same documentation standard as every other project in the series: full logic, I/O documentation, a live HMI, functional test procedures, and an honest account of what separates a strong simulation from a validated production system.

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End of document.